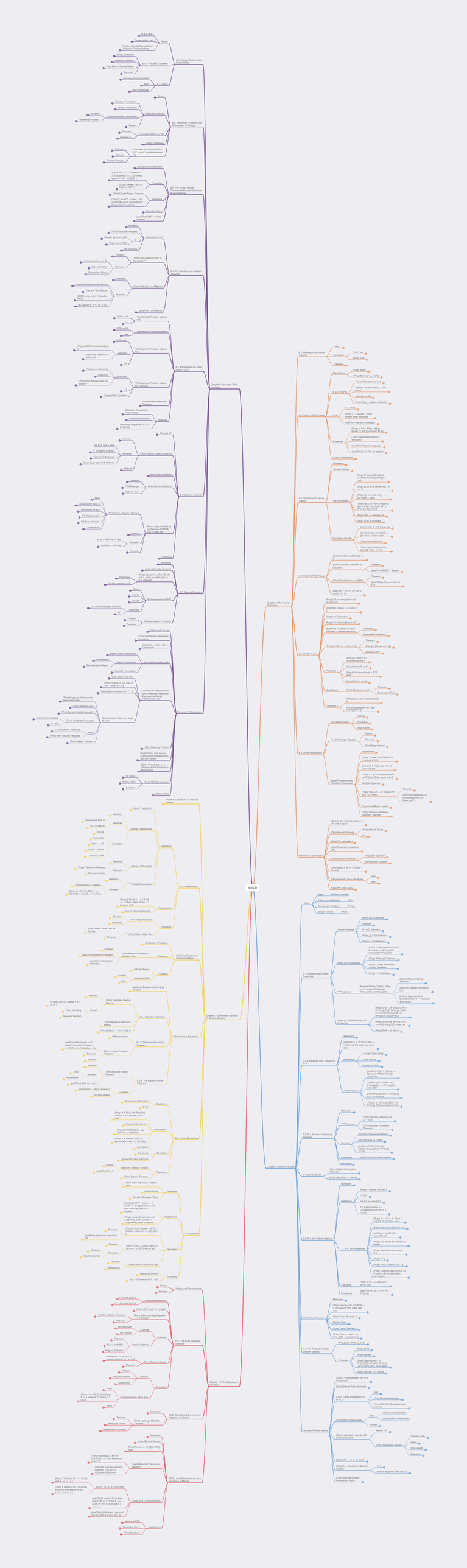

M383D

- Chapter 6: The Fourier Transform
  - 6.1: Motivation for Fourier Analysis
    - History
    - Motivation
      - Finite Heat
      - Infinite Heat
    - Core Idea
  - 6.2: The L^1(R^d) Theory
    - Plane Wave
      - Plane Wave
      - (Prop) MC/exp/ -Δ=l\xil^2
    - F on L^1(R^d)
      - Fourier Transform on L^1
      - (Prop) F \in B(L^1(R^d), L^\inf (R^d))
      - Example on \Xi
      - (Prop) \tau_y, dialtion, reflection
    - C_v
      - C_v (R^d)
      - (Prop) C_v \subset L^\infty closed linear subspace
      - (preThm) Riemann-Lebesgue
    - Properties
      - (Prop) In L^1, <Ff,g>=<f,Fg>, f*g\in L^1, F(f*g)=(2π)^d/2 Ff Fg
      - (Thm) Generalized Young's Inequality
      - (postThm) Young's Inequality
      - (postThm) (L^1, *) is an algebra
    - (Thm) Paley-Wiener
  - 6.3: The Schwartz Space Theory
    - Motivation
    - Schwartz Space
    - [3+2]Properties
      - (Prop) D \subset S \subset L^1(R^d) n L^\infty (R^d) [=> L^p]
      - (Prop) pf, gf, D^a f linear+cts : S —> S
      - (Prop) f_j --> f in S => f_j —> f in L^p for [1,\infty]
      - (Thm) f\in S => F\in C^\inf(R^d) with 1. F(D^a f) = (i\xi)^a Ff 2. D^a(Ff) = F[(-ix)^a f]
      - (Prop) \rho_n ~ \omega_ab
      - (Prop) (S,d) is complete
    - [1+C]Main Results
      - (preThm) F: S—>S linear+cts
      - (preThm) \phi = e^lxl^2/2 => \phi \in S + F\phi = \phi
      - (Thm) Plancherel on S
      - (Thm) f,g\in S => f*g \in S + (2π)^d/2 F [fg] = Ff Fg
  - 6.4: The L^2(R^d) Theory
    - (postThm) Parseval Identity on S
    - (Thm) Extension Theorem on D-u.cts-C
      - Theorem
      - (postThm) X NLS Y Banach
    - (Thm) Plancherel on L^2(R^d)
      - Theorem
      - (postThm) <Ff,g>=<f,Fg> on L^2
    - (preThm) F on L^2 (L^1nL^2 and L^2\L^1)
  - 6.5: The S' Theory
    - (Prop) i: D \xhookrightarrow S cts; cl(D)=S
    - (postThm) conv in D => conv in S
    - Tempered Distribution
    - (Prop) i: S' \xhookrightarrow D'
    - (postThm) S' \subset D' strict subspace + unique extension
      - Corollary
      - Example D^a \delta_0
    - (Thm) u \in S' <=> u \rho_n bdd
      - Theorem
      - Example [Tempered L^p]
      - Example D\S'
    - Properties 1
      - (Prop) [1,\infty], L^p \xhookrightarrow S'
      - (Prop) Define F on S'
      - (Prop) D^a Associativity + S'*S \in S'
      - (Prop) F/F^-1.. on S'
    - Main Result
      - (Thm) Plancherel on S'
        - Theorem
        - Example on F(1)
    - Properties 2
      - (Prop) \tau_y/D^a inside/outside F
      - (Prop) Associativity on * and F(S'*S)/FS'*FS
  - 6.6: Some Applications
    - The Heat Equation
      - Setting
      - F to solve
      - Heat Kernel
    - The Schrödinger Equation
      - Setting
      - F to solve
      - Schrödinger Kernel
    - Signal Processing and Translation Invariance
      - Signal/Filter
      - (Prop) T linear,TI => T[(e^\xi t)] = \psi(\xi) e^\xi t
      - (preThm) T linear, cts, TI => T D^a invariant
      - (Thm) T: L^p—>L^q linear,cts,TI => T(S)= u*(S) for some u \in S'
      - Multiplier Operator
      - (Thm) T(S)=u*S => T bdd in L^2 <=> Fu L^\infty
        - Theorem
        - (postThm) Multiplier <=> Convolution on S or T linear,cts,TI
      - Causal filter/Band Limited
      - (Thm) Shannon-Whittaker Sampling Theorem
  - Homework Observation
    - (Obs) [1,2] => F[L^p] is dense in L^q with 2 equal
    - (Obs) Hausdorff-Young
      - Generalizaed Young
      - HY
    - (Obs) F[f] = \lambda f
    - (Obs) Taylor's Formula from infty
    - (Obs) Variants of Waves
      - Telegraper Equation
      - Klein-Gordon Equation
    - (Obs) \delta_0 \in (H^k (R^d))^* for d<2k
    - (Obs) linear cts TI <=> Multiplier
      - HW
      - FNO
    - (Obs) H^k norm equiv
- Chapter 7: Sobolev Spaces
  - History
    - Italy
      - Dirichlet Principle
    - Hilbert and Göttingen
      - H^s
    - Murray and Nikodym
      - W^m,p
    - Sergei Sobolev
      - Math
  - 7.1: Definitions and Basic Properties
    - Sobolev Spaces
      - W^{m,p}(O) Definition
      - Example
      - H^m(O) Definition
      - W^{m,p}_0 (O) Definition
      - W^{-m,p} (O) Definition
    - W^{m,p}(O) Properties
      - (Prop) 1. W^{0,p}(O) = L^p(O) 2. m≥k≥0 => W^{m,p}(O) \xhookright W^{k,p}(O)
      - (Prop) W^{m,p}(O) Banach
      - (Prop) [1,infty) Separable, (1,infty) Reflexive
      - (Prop) H^m(O) Hilbert
    - ****Theorems
      - [Meyers-Serrin] (Thm) [1,\infty) => cl(C^\infty (O) \bigcap W^{m,p}(O)) = W^{m,p}(O)
        - [Global Approximation] Theorem
        - (preThm) Mollifier 3 Props on L^p
        - [Interior Approximation] (postThm) \phi * f —> f interior W^{m,p}(O')
    - W^{m,p}_0 (O)/W^{-m,p} (O) Properties
      - (Prop) [] => 1. W^{m,p}_0 (R) = W^{m,p} (R) 2. W^{m,p}_0 (O) \xhookright W^{m,p} (O) 3. W^{0,p}_0 (O) = L^p(O)
      - (Prop) [] => D^a: W^{m,p} (O) —> W^{m-lal,p} (O) linear cts
      - (Prop) Split +- of above
  - 7.2: Extensions from \Omega to R^d
    - Motivation
    - (preThm) [] E: W^{m,p} (R+) —> W^{m,p} (R) linear bdd norm-equi
    - Definitions
      - Locally C^{m,1}-bdry
      - C^{m,1}-bdry
      - Partition of Unity
    - ****[] Theorem
      - (preThm) C^{m-1,1}-bdry => {\phi_j} LFPOU st ll\phi_jll_{m,p} bdd
      - (Thm) C^{m-1,1}-bdry => E: W^{m,p}(O) —> W^{m,p}(R) linear bdd
      - (postThm) Lipschitz => W^{m,p}(O) ~ W^{m,p}(R)
      - (Thm) E_0: W^{m,p}_0 (O) —> W^{m,p} (R) linear bdd isometry
  - 7.3: The Sobolev Embedding Theorem
    - Motivation
    - ****Theorems
      - (Thm) Sobolev Inequality on C^1_0(R)
      - (Thm) Sobolev Embeding Theorem
    - Lemmas
      - (preThm) Generalized Hölder
      - (preThm) p>d => L^\infty
      - (preThm) [1,d) d (d,\infty) Sobolev Inequality on W^{m,p}_0 (O)
    - Corollaries
      - (postThm) Poincaré-Friedrichs
    - Summary
  - 7.4: Compactness
    - (Thm) Rellich-Kondrachov Theorem
    - (postThm) Weak => Strong
  - 7.5: The H^s Sobolev Spaces
    - Motivation
    - Definitions
      - Bessel Potential of Order s
      - H^s(R)
      - H^s(O) \iso H^s(R)/Z
      - Z + Lipschitz-bdry => Consistency on H^m(O) = H^s(O)
    - ****Λ^s / H^s Properties
      - (Prop) S'—>S' => 1. Λ^s+t = Λ^s Λ^t, 2. Λ^s^-1 = Λ^-s
      - (Prop) llull_H^s = llΛ^s ull_L^2
      - (preThm) (1+x^2)^s/2 ~ \sum_0^s x^k
      - (Prop) S is dense and H^s(R) is Hilbert
      - (Prop) s≥t => H^s \xhookright H^t
      - (Prop) H^-s
      - (Prop) H^s(O) Hilbert, llEll ≤ C
      - (Prop) Lipschitz-bdry, m \in Z => H^m(O) = W^{m,2}(O) with norm-equiv
    - Theorems
      - (Thm) m \in Z => H^m (R) = W^{m,2}(R)
    - Corollaries
      - (postThm) m \in Z => H^m = W^{m,2}
  - 7.6: A Trace Theorem
    - Motivation
    - (Thm) 0<k<d => R: H^s(R^d) —> H^{s-k/2}(R^d-k) linear bdd onto
    - (Thm) Trace Theorem 1
    - Normal Trace
    - (Thm) Trace Theorem 2
    - (Thm) C^{m-1,1}-bdry => H^m_0(O) = ker(\gamma)
  - 7.7: The W^{s,p}(\Omega) Sobolev Spaces
    - W^{s,p}(O) / W^{s,p}_0 (O)
    - Properties
      - (Prop) Norm
      - (Prop) Banach
      - (Prop) Lipschitz-bdry => W^{s,2}(O) = H^s(O), W^{s,2}_0(O) = H^s_0(O), norm-equiv
      - (Prop) SET/RCT/TC Holds
  - Homework Observations
    - (Obs) L^p interpolation and H^s interpolation
    - (Obs) Refined Trace Estimate
    - (Obs) Traces and Mean-0 for H^m_0
      - HW
      - (Thm) Poincare-Wirtinger
      - (Thm) PW+PF=Bramble-Hilbert Lemma
    - (Obs) POU Constructions
      - Bdd
        - Annular Decomposition
        - Subordinate Compactness
      - Unbdd
    - (Obs) \nabla (lul^r u) under RK costs integrability
      - Idea in HW
      - (Thm) Nemytskii Operator
        - Definition Wiki
        - Setup
        - Poly Growth
        - Examples
    - (Obs) ERT => E is cts if d<2
    - (Obs) H_w Spaces and Banach Algebra
      - H_w
      - Banach Algebra under \star_w
    - (Obs) Discrete Sobolev Inequality in Matrix
- Chapter 8: Boundary Value Problems
  - 8.1: Second Order Linear Elliptic PDEs
    - Settng
      - SOLE PDE
      - Conservation Law
      - Positive Definite/Semidefinite/ Uniformly Positive Definite
    - 8.1.1: Practical Examples
      - Heat Conduction
      - Electrical Potential
      - Fluid Flow in Porous Media
      - Summary
    - 8.1.2: BCs
      - Boundary Decomposition
      - BVP
      - Well-Posedness
  - 8.2: Variational Problems and Minimization of Energy
    - Setup
    - Regularity Varying
      - Classical Formulation
      - Strong Formulation
      - Variational/Weak Formulation
        - Analysis
        - Variational Problem
      - Remark
    - (Thm) H^1 BVP <=> VP
      - Theorem
      - Remark cu
    - Energy Functional
    - (Thm) add SPD + c≥0 => H^1 BVP <=> VP <=> Minimization of J
      - Theorem
      - Gateaux
      - Dirichlet Principle
  - 8.3: The Closed Range Theorem and Linear Operators Bounded Below
    - Orthogonal Complement
    - Properties
      - (Prop) NLS 1. Z^⊥ closed in X; 2. Z \subset Z^⊥⊥; 3. Z closed sub + X = X^** => Z=Z^⊥⊥
      - (Prop) A linear + cts => R(A)^⊥=N(A^*)
    - Theorems
      - (Thm) Closed Range Theorem
      - (Thm) X, Y=Y^**, A linear + cts => A bddbl <=> A injective R(A) Closed R(A)^⊥=N(A^*)
    - Bounded Below
    - (postThm) OMT => A cts inverted
  - 8.4: The Babuška-Lax-Milgram Theorem
    - Motivation to EU
      - Problem
      - (Thm) Poincaré Inequality
      - B
        - Bilinear Symmetric ≥0
        - Upper Lower bdd
      - VP 2nd Form
    - (Thm) Uniqueness of BVP of Standard H^1
      - Theorem
      - Remarks
        - nonunique of u=w+u_D
        - trace estimates
        - Generalized Riesz
    - (Thm) Babuška-Lax-Milgram
      - Theorem
      - Remarks
        - Ladyzhenskaya-Babuška-Brezzi
        - Trial and Test Spaces
        - [MOT] Leray-Lions / Browder-Minty
        - cts + linear S: Y^* x \X —> \X
    - (postThm) Lax-Milgram
  - 8.5: Applications to Linear Elliptic PDEs
    - The Dirichlet Problem without b(x)
      - BVP<=>VP
      - LM
    - The General Dirichlet Problem
      - BVP<=>VP
      - LM+
    - The Neumann Problem without b(x)
      - BVP<=VP
      - Remarks
        - VP gives ONLY weak as bdry in D'
        - Regularity Upgrade for BVP=>VP
      - LM
    - The Neumann Problem without b(x) and c(x)
      - BVP<=VP
        - Problem on Coercivity
        - \tilde{H}^1
        - (Thm) Poincaré Inequality on \tilde{H}^1
      - LM
      - Compatibility Condition
    - (Thm) Elliptic Regularity Theorem
    - Remarks
      - Ellipticity / Parabolicity / Hyperbolicity
      - Solvability Hierarchy
      - Regularity Upgrading for ALL EPH/TCD
  - 8.6: Galerkin Methods
    - Abstract VP
    - (Thm) Bubnov-Galerkin Method
      - Theorem
      - Remarks
        - for all n Trial = Test
        - H_n ranking + dense
        - Galerkin Orthogonal
        - [Céa] Quasi-optimal & Optimal
      - Method
    - Ritz-Galerkin Method
    - Finite Element Methods
      - Definition
      - FEM Example
      - FEM P1 Error
    - Petrov-Galerkin Method \ref{Brenner MTFEM} \ref{JTOden FA}
      - (Thm) Petrov-Galerkin Method
        - BLM
        - Discretized U_hxV_h
        - Discretized inf-sup
        - PG Orthogonality
        - PG Error Estimate
        - Convergence
      - Method
      - Remarks
        - for all n Trial = or != Test
        - coercive —> inf-sup
      - Example
    - Summary
  - 8.7: Green's Functions
    - Motivation
    - Green's Function for (L,B)
    - (Prop) G(.,y): O—>R is F.S. of L with L_x G(x,y)=\delta_0(x-y) B_x G(x,y)=0
      - Proposition
      - E + bdry corrector = G
    - Nonhomogeneous BVP
      - Setup
      - Implicit
      - Explicit
      - Examples
        - DP: Poisson Integral Formula
        - NP
    - Modified Green's Function
      - Problem
      - Definition
  - Homework Observations
    - (Tech) cu^k \in L^p
    - (Obs) Coordinates dependent Regularity
    - (Tech) cts => Ker = {0} => Closed sub
    - (Format) Formulating VP
      - Higher Order Formulation
      - Mixed Formulation
        - Lax-Milgram
        - Babuška-Lax-Milgram
      - Coupled Formulation
    - (Theory) On Inequalities of Korn, Friedrichs, Magenes-Stampacchia-Nečas and Babuška-Aziz
      - Setting Sym and Skw
      - (Thm) ll\nabla ull_2 = lldiv ull ^2 + llcurl ull_2^2
      - Setting Decomposition of H^1_0
      - Closed-Range Theory for grad-div-curl
        - (Thm) Magenes-Stampacchia-Nečas Inequality
        - (Thm) Babuška-Aziz
        - (Thm) General Necás Inequality
        - (Thm) Friedrichs Inequality
          - Harmonic Conjugates
          - F = BA
        - Korn
          - ***(Thm) Korn's Inequality
          - (Thm) Korn Second Inequality
        - (Thm) Master Theorem
    - (Obs) Elasticity Problem
    - (Tech) Trial + Test Spaces Construction by Weak Form and bdry Needs
    - (Tech) Perturbation <—> Change of Test functions in Mixed Form
    - (Tech) FEM Construction
      - VP [BSIL]
      - [MSEL] FEM
      - Summary
    - (Tech) Find G
- Chapter 9: Differential Calculus in Banach Spaces
  - Problem: Derivatives on Banach Spaces
  - 9.1: Differentiation
    - Definitions
      - llf(h)ll_Y=o(llhll_X)
      - Fréchet Differentiable
        - Definition
        - Remarks
          - Generalized from R
          - Df(x) \in B(X,Y)
        - Examples
          - f(x)=Ax
          - f(x)=(x,Ax)
          - f: R^n —> R
          - f: R^n —> R^m
          - f: (R^d)^p —> R
      - Gâteaux Differetiable
        - Definition
        - Remarks
          - (Prop) Fréchet => Gâteaux
          - Counterexample
      - *****Partial Differetiations
        - Definition
        - Remarks
          - Partial exists \=> Gâteaux
          - (Prop) D_j F(x) \in B(U,Y) => Df(x_0) h = \sum D_j F(x_0) h_j
    - Propositions
      - (Prop) f: U\sub X —> Y F-diff. => 1. Df(x) unique cts on U 2. Linearity of D
      - (preThm) Local-Lipschitz
    - Theorems
      - *****(Thm) Chain Rule
        - Theorem
        - Examples
      - *****(Thm) Mean Value Thm
        - (Prop) Mean Value Thm for Curves
        - Theorem
  - 9.2: Fixed Points and Contractive Maps
    - Contraction / Fixed pts
    - Theorems
      - (Thm) Banach Contraction Mapping Thm
        - Theorem
        - (postThm) Fixed Point Iteration
        - (postThm) Contraction Detection
    - Examples
      - IVP with Picard
      - Nonlinear PDE
        - Solution
        - Why
  - 9.3: Nonlinear Equations
    - Nonlinear Equations Solving by Iteration
    - 9.3.1: Newton's Methods
      - (Thm) Simplified Newton Method
        - Theorem
        - Remark
          - B_\delta (f(x_0)) \subset f(B_r (x_0))
          - Fixed pt setting
          - Newton's Iteration
      - (Thm) Newton-Kantorovich Method
      - (Thm) BCBC => f^1\in C^{0,1}
    - 9.3.2: The Inverse Function Theorem
      - Diffeomorphism
      - (Thm) Inverse Function Theorem
        - (preThm) X,Y Banach => 1. GL(X,Y) \sub B(X,Y) open 2. J_XY (A) = A^-1 bijective + cts
        - Theorem
        - Remark
    - 9.3.3: The Implicit Function Theorem
      - (Thm) Implicit Function Theorem
        - Theorem
        - Remarks
          - Z={0}
          - Visualization
        - (postThm) Slice on y=y_0
      - Examples
        - p(A,\lambda) = det(A-\lambda I)
        - IVP Perturbation
  - 9.4: Higher Derivatives
    - Defintions
      - \B^n(X,Y) and B^n(X,Y)
      - D^n f
    - Propositions
      - (Prop) X,Y NLS, f\in \B^n(X,Y) => f cts <=> f cts at x_0 <=> f bdd
      - (Prop) llfll_B^n(X,Y)
      - (Prop) B^k(X,B^l(X,Y)) \iso B^k+l (X,Y) Banachly
      - (Prop) X,Y Banach, f\in C^2, k+l=n => D^n f(x) = D^kD^l f(x)
    - Examples
      - A\in B(X,Y)
      - f(x)=(x,Ax)
      - F(g)(x)=\int K(x,y) g^p(y) dy
    - Theorems
      - (preThm) Schwarz Lemma
        - Lemma
        - (postThm) D^n f
      - (Thm) Taylor's Theorem
  - 9.5: Extrema
    - Defintions
      - min / max / extremum / relative / strict
      - Critical Points
      - Convex / Concave / Strict
    - Propositions
      - (Prop) f,g \in X^* , a,b>0 => 1. convex => af+bg convex 2. one strict => af+bg strict 3. -f concave
      - (Prop) convex U convex f =>1. supporting plane 2. strict => s.supporting plane 3. Df(x)=0
    - Theorems
      - (Thm) X NLS, U open, f \in C^1, Relative extremum => Df(x_0) = 0
        - Theorem
        - (postThm) extremum are critical pts
      - (Thm) X NLS, U open, f \in C^2 at r.min x => D^2f(x)(h,h) ≥ 0
        - Theorem
        - Remarks
          - Relations
          - Counterexample
      - (Thm) Second Derivative Test
        - Theorem
        - Remark:IPS
    - Examples
      - Quadratic function
      - J(v) = 1/2 ll\nabla vll^2 - (f,v)
- Chapter 10: The Calculus of Variations
  - History and Development
    - History
    - Problem
  - 10.1: The Euler-Lagrange Equations
    - Admissible Variations
      - C^1_0([a,b];R^d)
      - C^1_{\a,\b}([a,b];R^d)
    - (Thm) f C^1 => F C^1 and DF
    - (Thm) Euler-Lagrange Equation <=> Critical pts
      - (preThm) Dubois-Reymond
      - Theorem
    - Extremal
      - Example
        - Shortest Path
        - No Solution
      - Regular Extremal
        - Extremal
        - EL => 2nd ODE
        - Regular Extremal
    - (Thm) Beltrami Identity
      - (Prop) f \in C^2, y \in C^1 regular extremal => f \in C^2
      - Theorem
    - Examples
      - Catenary
        - Problem
        - Regular Extremal
        - Goldschdmit
      - Brachistochorne with 1 free
        - Find f
        - (Thm) y \in C^2_{a,*} minimize F => y satisfies EL and D_3 f (b)=0
        - Solve
  - 10.2: Constrained Extrema and Lagrange Multipliers
    - Motivation
    - (Thm) Lagrange Multiplier Theorem
      - Theorem
      - Method of Search
      - Isopetrimetric Problem
  - 10.3: Lower Semicontinuity and Existence of Minima
    - Motivation
    - Lower Semicontinuous
    - (Prop) f l.s.c <=> f^{-1}[(a,\infty]] open
    - Direct Method of Calculus of Variation)
      - (Thm) M compact, f: M—>(-inf,inf] l.s.c. => f bdd below and takes min
      - (Thm)] M w.closed sub of X r.Banach, f w.l.s.c => f(x0)=min_M f(x)>-inf
    - To get w.l.s.c. and commons
      - w.l.s.c. <=> s.l.s.c + convex
        - (Prop) X Banach, f:X—>(-inf,inf] w.l.s.c. => f s.l.s.c.
        - (Thm) X Banach, f:X—>(-inf,inf], V={f<inf}, f convex on V and s.l.s.c. => f w.l.s.c.
      - (preThm) f convex, \O domain, p\in [1,inf] => F: L^p(\O) —> [0,inf] by \int_\O f(u(x))dx is s.+w.l.s.c.
      - (postThm) \O domain, 1≤p,q<inf => L^q norm is w.l.s.c. on L^p
    - Applications
      - Nonlinear PDE
      - Rectifiable Curve
      - (Thm) Geodesic